

Delooping the Connecting Maps in the Taylor Tower

Greg Arone — January 18, 2007. Joint work with W. Dwyer and K. Lesh.

Transcribed from three-page handwritten notes. [Download the printable transcript \(PDF\)](#). Arrow diagrams are also described in words. Abbreviated notation in the notes is retained.

Page 1: Taylor towers and connecting maps

Given X , get a tower of fibrations. This gives a way to pass from stable to unstable information. The Taylor tower of the identity has compatible maps from X to every stage:

$$\cdots \rightarrow P_3(X) \rightarrow P_2(X) \rightarrow QX = \Omega^\infty \Sigma^\infty X.$$

For good spaces,

$$\lim_n P_n(X) = X.$$

We know a lot about this:

$$D_n(X) = \text{Fiber}(P_n(X) \rightarrow P_{n-1}(X)) \simeq \Omega^\infty((W_n \wedge X^{\wedge n})_{h\Sigma_n}).$$

The annotations identify W_n as the derivative, $X^{\wedge n}$ with the x^n term, and homotopy orbits with the $1/n!$ term of a Taylor series.

Since these are infinite loop spaces, there are maps

$$D_n(X) \rightarrow P_n(X) \rightarrow P_{n-1}(X) \rightarrow BD_n(X).$$

Thus there are connecting maps, in the style of k -invariants,

$$D_{n-1}(X) \rightarrow BD_n(X).$$

These give the d_1 differential in the spectral sequence for $\pi_*(X)$ using the tower.

Fix a prime p and let $X = S^m$: m odd when $p > 2$, and any m when $p = 2$.

Theorem. In this case $D_n(X) \simeq *$ unless $n = p^k$.

We can then regrade the tower. Put $DI_k = D_{p^k}(X)$. The diagram is

$$\cdots \rightarrow P_{p^2}(X) \rightarrow P_p(X) \rightarrow P_1(X) = QX,$$

with fiber maps $DI_2 \rightarrow P_{p^2}(X)$ and $DI_1 \rightarrow P_p(X)$.

For the connecting map, use the equivalence $P_{p^{k-1}}(X) \simeq P_{p^{k-1}}(X)$ and the map from DI_{k-1} into the latter stage. Composing with the map to BDI_k gives the connecting map labeled α_{k-1} :

$$\alpha_{k-1} : DI_{k-1} \rightarrow BDI_k.$$

There is another functor, from complex inner-product vector spaces to spaces,

$$V \mapsto BU(V).$$

There is a Weiss calculus starting with

$$BU = \varinjlim_V BU(V).$$

After p -completing, there is again a faster collapse.

Page 2: Deloopings and filtrations

The tower for $BU(V)$ has compatible maps to

$$\cdots \rightarrow T_p(V) \rightarrow T_{p^0}(V) \rightarrow T_0(V) = BU,$$

where T_0 is the constant term. The derivatives are written in the notes as

$$DU_k(V) \simeq \Omega^\infty((\underline{W}_{p^k} \wedge S^{p^k V})_{hU(p^k)}).$$

Here the first factor is annotated as a $U(p^k)$ -spectrum, and $S^{p^k V}$ as the one-point compactification $(\mathbb{C}^{p^k} \otimes V)^+$.

Again there are connecting maps

$$\beta_k : DU_k \rightarrow BDU_{k+1}.$$

These are not infinite loop maps, but:

Theorem. α_k and β_k are k -fold loop maps. Thus, for example, there exists

$$\tilde{\alpha}_k : B^k DI_k \rightarrow B^{k+1} DI_{k+1}, \quad \alpha_k \simeq \Omega^k \tilde{\alpha}_k.$$

Example: S^1 . Even in this case, α_k is not an infinite loop map. The diagram has $S^1 \rightarrow P_p(S^1)$ and $S^1 \rightarrow QS^1$, the map $P_p(S^1) \rightarrow QS^1$, the fiber map $DI_1 \rightarrow P_p(S^1)$, and the connecting map $QS^1 \rightarrow BDI_1$.

Using the deloopings gives a sequence of spaces

$$S^1 \rightarrow QS^1 \rightarrow BDI_1 \rightarrow B^2 DI_2 \rightarrow \cdots.$$

The forward maps labeled α_k are not infinite loop maps. The backward maps f_k , drawn as dashed arrows in the notes, are infinite loop maps coming from

$$\mathrm{Sp}^\infty(X) \simeq K(H_*X).$$

The inclusions

$$S^0 \rightarrow \mathrm{Sp}^2(S^0) \rightarrow \cdots \rightarrow \mathrm{Sp}^\infty(S^0) = H\mathbb{Z}$$

give a filtration of $H\mathbb{Z}$. Also, p -locally,

$$\mathrm{Sp}^n(S^0)/\mathrm{Sp}^{n-1}(S^0) \simeq *$$

unless $n = p^k$. One also gets

$$H\mathbb{Z} \leftarrow S^0 \leftarrow \Sigma^{-1}(\mathrm{Sp}^p(S^0)/\mathrm{Sp}^1(S^0)) \leftarrow \cdots.$$

Looping the other sequence gives

$$\mathbb{Z} \leftarrow QS^0 \leftarrow \Omega BDI_1 \leftarrow \cdots,$$

which is Ω^∞ of the tower above. The term ΩBDI_1 is annotated $QB\Sigma_p$.

There is an analogous filtration in the orthogonal case:

$$bu \rightarrow A_1 \rightarrow \cdots \rightarrow H\mathbb{Z}.$$

Page 3: The conjecture and a proof idea

The Weiss calculus gives

$$BU(1) \rightarrow BU \rightarrow BDU_0 \rightarrow B^2DU_1 \rightarrow \cdots,$$

where $BU(1) = \mathbb{CP}^\infty$ and $BDU_0 = Q\Sigma^2\mathbb{CP}^\infty$.

Taking Ω^2 gives

$$\mathbb{Z} \rightarrow \mathbb{Z} \times BU \rightarrow Q\mathbb{CP}^\infty \rightarrow \Omega^2 B^2DU_1 \rightarrow \cdots.$$

There are forward maps β_k and backward maps ∂_k . This is the same as the filtration from above.

Conjecture. There exist deloopings α_k, β_k that act as a contracting homotopy; for example,

$$f_k \circ \alpha_k + \alpha_{k+1} \circ f_{k+1} \simeq \mathrm{Id}.$$

Consequently the “chain complexes” are “exact,” and the homotopy spectral sequence collapses at E_2 .

Idea of proof. How can we show a map is a k -fold loop map? Given $\Omega^k X \rightarrow \Omega^k Y$, find the image of

$$\Omega^k : \mathrm{Map}(X, Y) \rightarrow \mathrm{Map}(\Omega^k X, \Omega^k Y).$$

These are too big. Instead look at natural transformations. The notes write the comparison as

$$\mathrm{Nat}(B^k(-), B^{k+1}(-)) \xrightarrow{\Omega^k} \mathrm{Nat}(DI_k(-), BDI_{k+1}(-)).$$

In the Weiss story, only consider $V \mapsto S^1 \wedge S^V$.

The last line singles out maps of the form

$$\Omega^d \Omega^\infty \Sigma^\infty (X^{\wedge 2^k})_{h\wr^k \Sigma_2} \rightarrow \Omega^d Q(X^{\wedge 2^{k+1}})_{h\wr^{k+1} \Sigma_2}.$$